DDS – DARK DIGITAL SPACE

CYBERSECURITY FUNDAMENTALS

Complete Student Training Manual

3-Month Program · 12 Weekly Modules · Beginner to Job-Ready

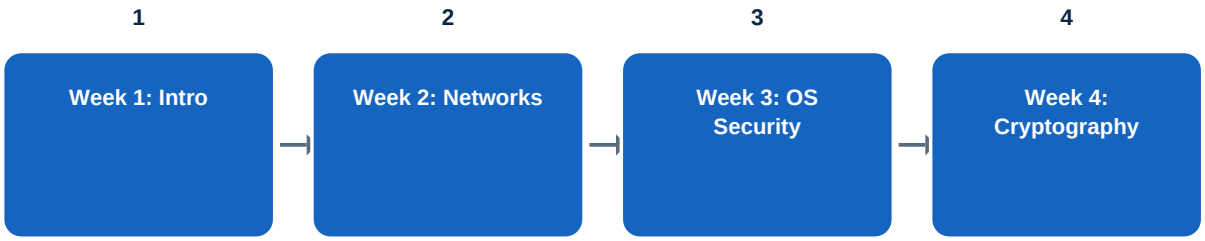
Duration	3 Months (12 Weeks)
Level	Beginner → Intermediate
Instructor	DDS Academy
Edition	2026

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Month 1 — Foundations

Month 1 builds the vocabulary and mental models every cybersecurity professional relies on: what we're protecting, how computers and networks actually work, how operating systems enforce boundaries, and how cryptography keeps information private and trustworthy.



Week 1

Introduction to Cybersecurity

Learning Objectives

- Define cybersecurity.
- Explain why cybersecurity is important.
- Identify different types of hackers.
- Understand the CIA Triad.
- Recognize common cyber threats.
- Apply basic online safety practices.

What is Cybersecurity?

Imagine you own a house. You lock your doors, close your windows, install cameras, and keep important documents in a safe. A computer works almost the same way — it stores valuable information that needs to be protected.

Cybersecurity is the practice of protecting computers, mobile devices, networks, software, and digital information from unauthorized access, theft, damage, or disruption. It also involves preventing accidental data loss, protecting privacy, and keeping systems running reliably.

Why Cybersecurity Matters

Every day, millions of people rely on the internet for banking, shopping, social media, business, government services, healthcare, and education. When these systems aren't properly protected, attackers can steal money, personal information, or disrupt essential services — for example, a hospital losing access to patient records because of ransomware, or a student's email being taken over due to a weak password.

The CIA Triad

The CIA Triad — Confidentiality, Integrity, and Availability — is one of the most important concepts in cybersecurity. Confidentiality means only authorized people can access information. Integrity means data stays accurate and unaltered. Availability means systems are accessible whenever authorized users need them.

Types of Hackers

White Hat hackers are ethical professionals authorized to test systems and find weaknesses before criminals do. Gray Hat hackers access systems without permission but usually don't intend harm — it's still illegal. Black Hat hackers break in illegally for personal gain, stealing money or data, or installing ransomware.

Strong Passwords & MFA

A strong password is at least 12 characters, mixes upper/lowercase letters, numbers, and symbols, avoids personal information, and is unique per account. Multi-Factor Authentication (MFA) adds a second layer of proof — like a code from your phone — so a stolen password alone isn't enough to break in.



Figure 1.1 — The CIA Triad

> **Scenario**
> A university student receives an email claiming to be from IT, asking them to “verify” their account through a link. The site looks genuine, so they log in — and later find their account compromised. What warning signs were missed, and how could MFA have helped?

Key Terms

Cybersecurity	**Confidentiality**
Integrity	**Availability**
Malware	**Phishing**
Multi-Factor Authentication (MFA)	**Ethical Hacker**

Review Questions

1. Define cybersecurity in your own words.
2. Explain the three principles of the CIA Triad.
3. Differentiate between White Hat, Gray Hat, and Black Hat hackers.
4. Why is multi-factor authentication important?

5 List five characteristics of a strong password.

Practical Activity

- Create a password that meets the strong password guidelines, and enable MFA on one personal account.
- Identify three examples of phishing emails from publicly available awareness materials and explain why they look suspicious.

Week 1

Supplementary Diagrams



Figure 1.2 — White Hat vs. Gray Hat vs. Black Hat

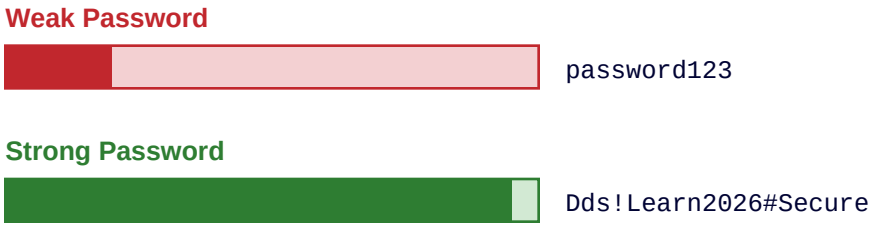


Figure 1.3 — Weak vs. Strong Password Strength

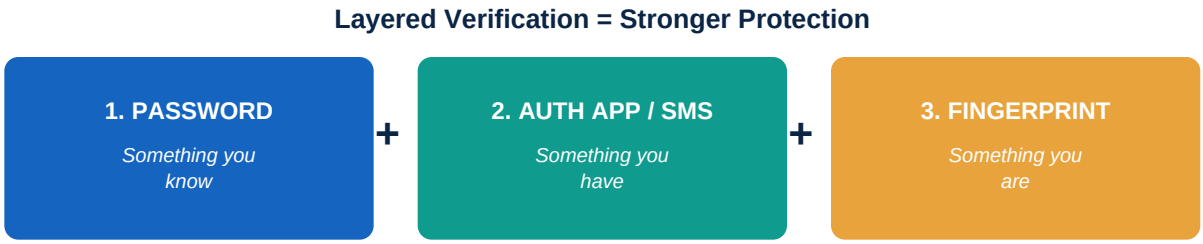


Figure 1.4 — Layers of Multi-Factor Authentication

Week 2

Computer & Network Fundamentals

Learning Objectives

- Explain how computers communicate over a network.
- Describe the client-server model.
- Identify common network devices and their roles.
- Understand IP addresses, MAC addresses, and DNS.
- Recognize common ports and protocols.

How Networks Move Data

A network is a group of devices connected so they can share information. When you load a website, your device (the client) sends a request across the network to a server, which sends the webpage back. This is called the client-server model, and it underlies almost everything you do online.

Data doesn't travel as one solid piece — it's broken into small units called packets, sent independently, and reassembled at the destination. This makes networks efficient and resilient, but it also means an attacker who intercepts traffic can potentially read or tamper with packets if they aren't encrypted.

Addressing: IP, MAC, and DNS

Every device on a network needs an address. An IP address (like 192.168.1.10) identifies a device on a network, similar to a postal address. A MAC address is a unique hardware identifier burned into a network card. DNS (Domain Name System) translates human-friendly names like example.com into IP addresses computers can route traffic to — acting like the internet's phone book.

Common Network Devices

- Router — directs traffic between networks (e.g., your home network and the internet).
- Switch — connects devices within the same local network.
- Firewall — filters traffic based on security rules (covered in depth in Week 5).
- Access Point — provides Wi-Fi connectivity to wireless devices.

Ports and Protocols

Protocols are agreed-upon rules for how data is formatted and exchanged — for example, HTTP/HTTPS for web browsing, and SMTP for email. Ports are numbered channels on a device that let different services run at once; port 443 is the standard port for secure HTTPS traffic, while port 80 is standard, unencrypted HTTP.

Simplified Network Model

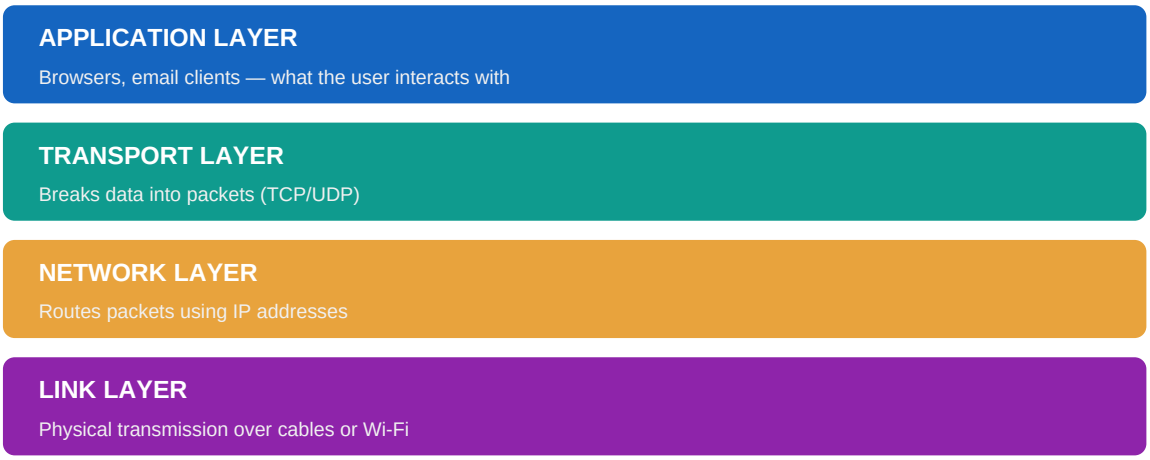


Figure 2.1 — Simplified Network Model (Application to Physical)

Key Terms

Network	**Client-Server Model**
IP Address	**MAC Address**
DNS	**Router**
Switch	**Protocol**
Port	**Packet**

Review Questions

1. Explain the client-server model with an everyday example.
2. What is the difference between an IP address and a MAC address?
3. What role does DNS play when you visit a website?
4. Name two common network devices and describe their function.
5. Why does port 443 matter for online safety?

Practical Activity

- Use a command-line tool (e.g., ipconfig/ifconfig) to find your device's own IP and MAC address.
- Trace the path of a webpage request: write out, step by step, what happens from typing a URL to seeing the page load.

Week 3

Operating System Security

Learning Objectives

- Explain why the operating system is a key security boundary.
- Configure basic user accounts and permissions.
- Explain the purpose of patch management.
- Recognize insecure default OS configurations.
- Understand the role of logging and monitoring.

Why the OS Matters

The operating system (Windows, macOS, Linux, etc.) sits between hardware and every application you run. It decides who can access which files, which programs can run, and what each user is allowed to do. If the OS is poorly configured, every application on top of it inherits that weakness.

Users, Groups, and Permissions

Modern operating systems separate accounts into regular users and administrators. The principle of least privilege means every account should have only the access it needs to do its job — a regular user account shouldn't have administrator rights for everyday tasks, since a compromised admin account gives an attacker far more control.

File permissions determine who can read, write, or execute a given file. On Linux, for example, permissions are shown as combinations like rwx (read, write, execute) for the owner, group, and everyone else.

Patch Management

Vendors regularly release updates (patches) that fix security vulnerabilities. Unpatched systems are one of the most common ways attackers gain access, because known vulnerabilities are publicly documented once a patch exists. A good patch management routine applies critical updates promptly, ideally through automated update policies.

Logging and Monitoring

Operating systems keep logs of events — logins, file access, errors, and more. Reviewing logs (or using automated monitoring tools) helps detect unusual activity early, such as repeated failed login attempts that might indicate a password-guessing attempt.

Where OS Security Sits

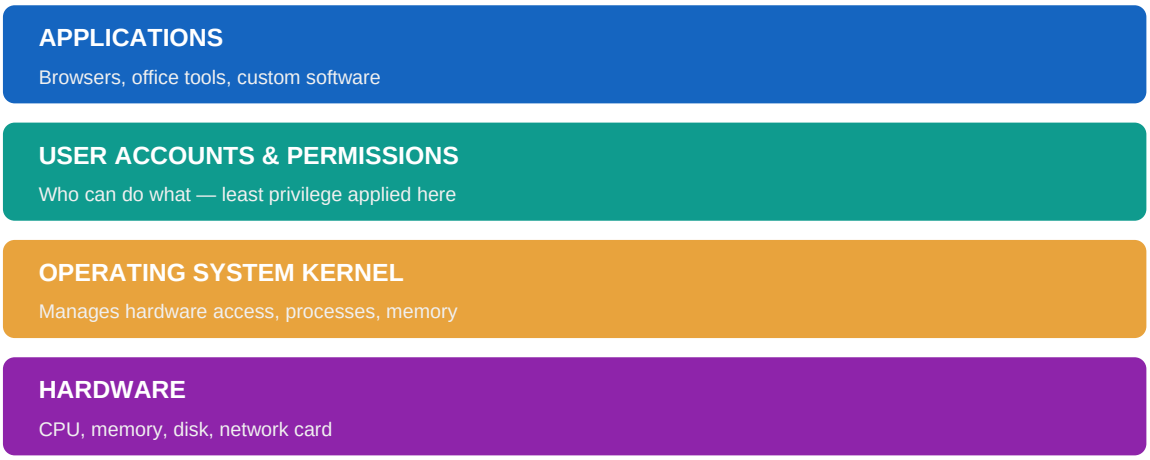


Figure 3.1 — Operating System Security Layers

Key Terms

Operating System	**Least Privilege**
File Permissions	**Patch Management**
Vulnerability	**Logging**
Administrator Account	**Hardening**

Review Questions

1. Why is the principle of least privilege important?
2. What is patch management, and why does it matter?
3. Give an example of a file permission and what it controls.
4. How can logs help detect a security incident?
5. What is the risk of using an administrator account for everyday tasks?

Practical Activity

- Review the user accounts on your own computer and identify which ones have administrator rights.
- Check whether your operating system has pending updates and describe what they address.

Week 4

Cryptography Basics

Learning Objectives

- Differentiate encryption from hashing.
- Explain symmetric vs. asymmetric encryption.
- Describe how HTTPS/TLS protects data in transit.
- Understand the purpose of digital signatures and certificates.

Encryption vs. Hashing

Encryption transforms readable data (plaintext) into unreadable data (ciphertext) using a key, and it's reversible — someone with the right key can decrypt it back. Hashing transforms data into a fixed-length value but is one-way: you can't reverse a hash back into the original data. Hashing is commonly used to store passwords safely, since the system only needs to compare hashes, never the original password.

Symmetric vs. Asymmetric Encryption

Symmetric encryption uses one shared secret key for both encrypting and decrypting — fast, but both parties need a secure way to share that key first. Asymmetric encryption uses a key pair: a public key anyone can use to encrypt, and a private key only the owner holds to decrypt. This solves the key-sharing problem and underlies much of secure web communication.

HTTPS and TLS

HTTPS is HTTP secured with TLS (Transport Layer Security). When you see the padlock icon in your browser, TLS has encrypted the connection between your device and the website, protecting data like passwords and payment details from being read if intercepted. TLS uses asymmetric encryption to safely establish a shared symmetric key for the rest of the session — combining the strengths of both approaches.

Digital Signatures and Certificates

A digital signature proves a message or file came from a specific sender and wasn't altered, using the sender's private key. Digital certificates, issued by trusted Certificate Authorities, bind a public key to an identity (like a website's domain), which is how your browser confirms it's really talking to the real example.com and not an impostor.

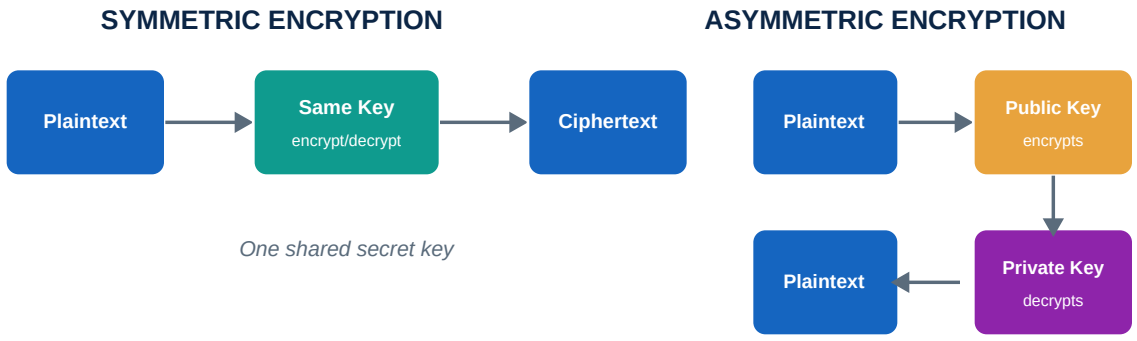


Figure 4.1 — Symmetric vs. Asymmetric Encryption

Key Terms

Encryption	**Hashing**
Symmetric Key	**Asymmetric Key**
TLS	**HTTPS**
Digital Signature	**Certificate Authority**

Review Questions

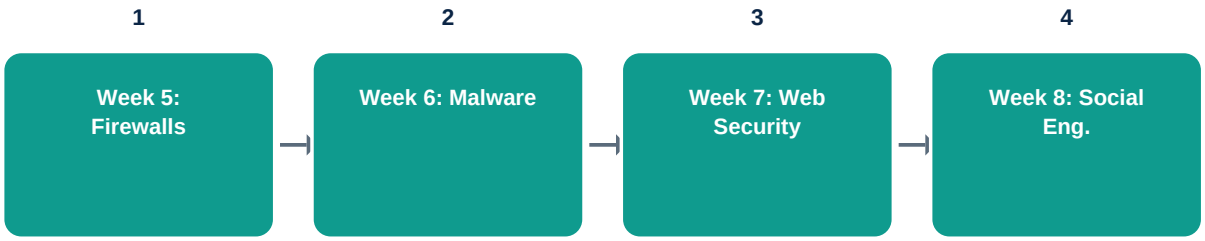
1. What is the key difference between encryption and hashing?
2. Why is asymmetric encryption useful for solving the key-sharing problem?
3. What does the padlock icon in a browser actually confirm?
4. What role does a Certificate Authority play?
5. Why are passwords typically hashed rather than encrypted?

Practical Activity

- Check a website you use daily for its certificate details (click the padlock icon) and note the issuing authority.
- In your own words, explain to a classmate why HTTPS matters for online banking.

Month 2 — Defense & Threats

Month 2 shifts from foundations to active defense: how networks are protected at the perimeter, how malicious software behaves, where web applications are commonly attacked, and how attackers manipulate people rather than machines.



Week 5

Network Security & Firewalls

Learning Objectives

- Explain how firewalls filter traffic.
- Describe common VPN use cases.
- Understand network segmentation.
- Identify signs of suspicious network traffic.

Firewalls

A firewall inspects incoming and outgoing traffic and allows or blocks it based on security rules — similar to a security guard checking IDs at a building entrance. Basic packet-filtering firewalls check simple criteria like source/destination address and port. Stateful firewalls go further, tracking the state of active connections to make smarter decisions about what traffic to allow.

VPNs (Virtual Private Networks)

A VPN creates an encrypted tunnel between a device and a remote network, so traffic traveling through it can't easily be read by someone else on the same network — useful on public Wi-Fi, or for remote employees securely reaching company resources.

Network Segmentation

Segmentation divides a network into smaller zones, so that if one part is compromised, the attacker can't automatically reach everything else. For example, a company might keep guest Wi-Fi, employee workstations, and servers holding sensitive data on separate network segments.

Monitoring for Anomalies

Intrusion Detection/Prevention Systems (IDS/IPS) watch network traffic for patterns that match known attacks or unusual behavior — like a sudden spike in traffic to an unfamiliar destination — and can alert administrators or automatically block it.

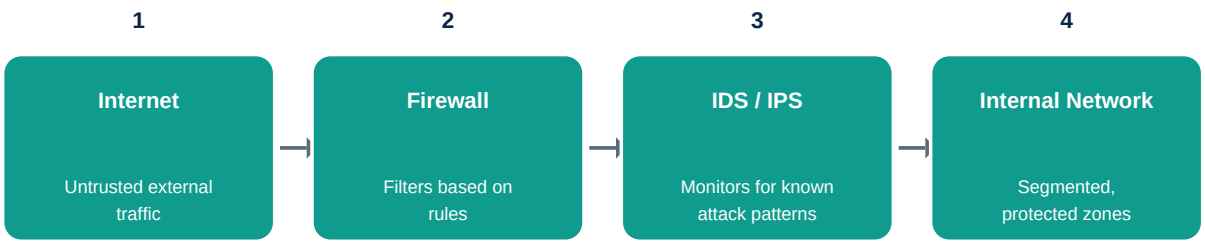


Figure 5.1 — Simplified Network Security Architecture

Key Terms

Firewall	**VPN**
Network Segmentation	**IDS**
IPS	**Packet Filtering**
Stateful Inspection	**Perimeter Security**

Review Questions

1. How does a stateful firewall differ from a basic packet-filtering firewall?
2. Why would a company use a VPN for remote employees?
3. Explain the benefit of network segmentation with an example.
4. What is the difference between an IDS and an IPS?

Practical Activity

- Check whether your home router has a built-in firewall and review its basic settings.
- Research one real VPN provider and summarize what encryption standard it advertises using.

Week 6 Malware Analysis Basics

Learning Objectives

- Classify common types of malware.
- Describe typical infection vectors.
- Explain the difference between static and dynamic analysis at a conceptual level.
- Understand the role of antivirus/EDR tools and indicators of compromise.

Types of Malware

Malware is malicious software designed to damage, disrupt, or gain unauthorized access to systems. Viruses attach themselves to legitimate files and spread when those files run. Worms self-replicate across networks without needing a host file. Trojans disguise themselves as legitimate software. Spyware secretly monitors user activity. Ransomware encrypts files and demands payment for the decryption key.

How Malware Spreads (Infection Vectors)

- Phishing emails with malicious attachments or links.
- Compromised or fake websites (drive-by downloads).
- Infected removable media, like USB drives.
- Exploiting unpatched software vulnerabilities.

Analyzing Malware (Conceptual Overview)

Security professionals study malware to understand and stop it. Static analysis examines a file without running it — looking at its code structure or known signatures. Dynamic analysis runs the file in an isolated, controlled environment called a sandbox to observe its behavior safely, without risking real systems.

Detection: Antivirus, EDR, and IOCs

Antivirus software scans for known malware signatures. Endpoint Detection and Response (EDR) tools go further, watching for suspicious behavior even from unknown threats. Indicators of Compromise (IOCs) — like unusual outbound connections or unexpected new files — help analysts recognize that a system may be infected.

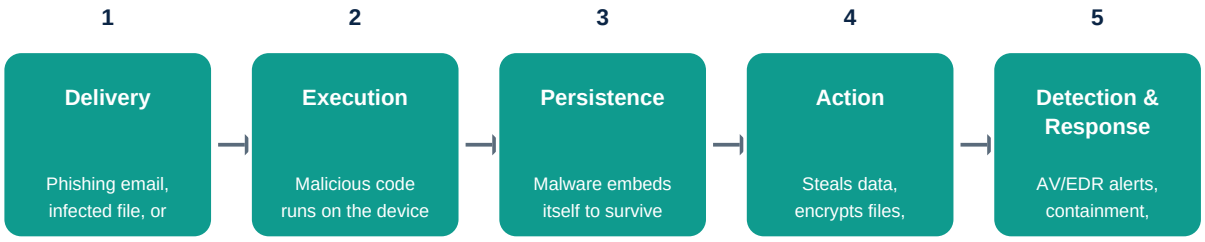


Figure 6.1 — Typical Malware Lifecycle

Key Terms

Malware	**Virus**
Worm	**Trojan**
Spyware	**Ransomware**
Sandbox	**Endpoint Detection and Response (EDR)**
Indicator of Compromise (IOC)	

Review Questions

1. What distinguishes a worm from a virus?
2. List three common ways malware infects a device.
3. What is a sandbox, and why is it used in malware analysis?
4. How does EDR differ from traditional antivirus?
5. Give an example of an Indicator of Compromise.

Practical Activity

- Research one well-known ransomware incident and summarize what happened, who was affected, and what lessons were learned.
- Review your own device's antivirus/security settings and confirm real-time protection is enabled.

Week 7

Web Application Security

Learning Objectives

- Describe the OWASP Top 10 at a conceptual level.
- Explain why input validation matters.
- Recognize insecure authentication and session practices.
- Understand the role of HTTPS in web application security.

Why Web Apps Are a Common Target

Web applications are exposed to anyone on the internet, making them an attractive target. The OWASP Top 10 is a widely referenced list of the most critical web application security risks, maintained by the Open Web Application Security Project, used by developers and security teams worldwide as a baseline checklist.

Common Risk Categories (Conceptual Overview)

- Injection — untrusted input is interpreted as commands instead of data.
- Broken Authentication — weaknesses in login/session handling let attackers impersonate users.
- Sensitive Data Exposure — data isn't properly protected in storage or transit.
- Security Misconfiguration — default settings, unnecessary features, or verbose error messages leave doors open.
- Cross-Site Scripting (XSS) — malicious scripts run in another user's browser.

Input Validation and Secure Design

Input validation means checking and sanitizing anything a user submits — form fields, URLs, uploaded files — before it's processed or stored. Treating all user input as untrusted by default is a foundational secure-coding principle, since it closes off many of the most common attack categories at the source.

Authentication and Session Security

Secure applications enforce strong password policies, support MFA, and issue session tokens that expire and can't easily be guessed or stolen. HTTPS protects these tokens and credentials from being intercepted while traveling across the network.

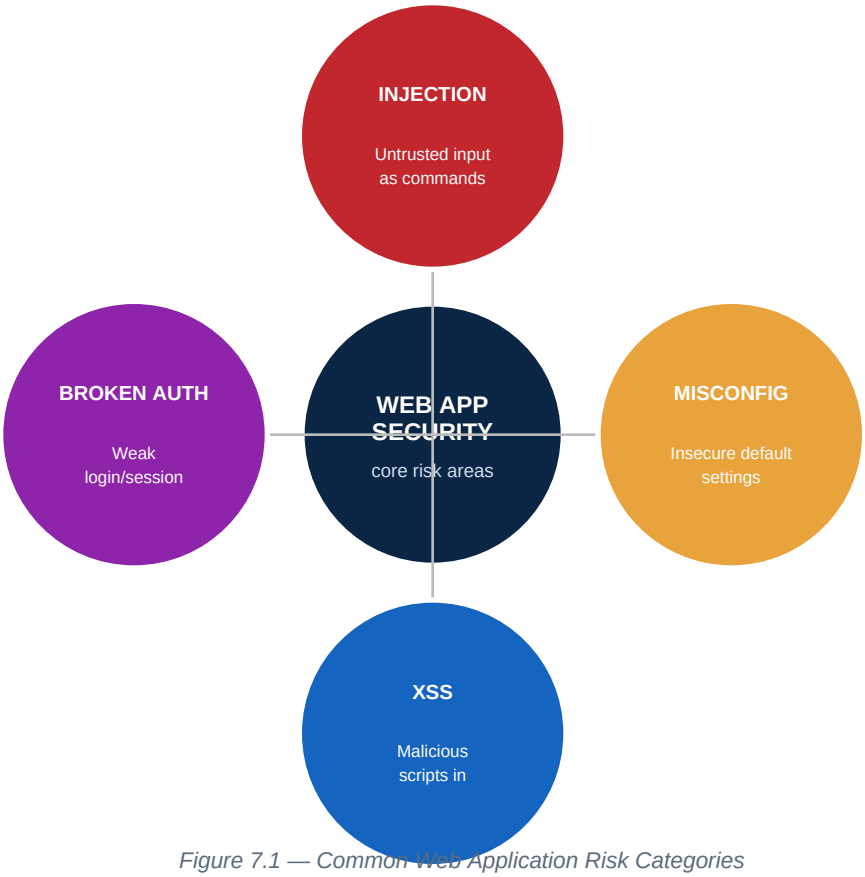


Figure 7.1 — Common Web Application Risk Categories

Key Terms

OWASP	**Injection**
Broken Authentication	**Cross-Site Scripting (XSS)**
Security Misconfiguration	**Input Validation**
Session Token	

Review Questions

1. What is the purpose of the OWASP Top 10?
2. Why is input validation considered a foundational security practice?
3. What risks fall under 'Broken Authentication'?
4. How does HTTPS support web application security?
5. Give an example of a security misconfiguration.

Practical Activity

- Look up the current OWASP Top 10 list online and compare it to the categories discussed in class.

- Review the login page of a site you use and note which security signals you can observe (HTTPS, MFA option, lockout after failed attempts).

Week 8

Social Engineering & Phishing Defense

Learning Objectives

- Identify common social engineering tactics.
- Recognize phishing red flags.
- Explain the concept of a 'human firewall'.
- Design basic security awareness practices.

Why Attackers Target People

Technical defenses can be strong, but people can still be tricked into bypassing them voluntarily. Social engineering manipulates human psychology — trust, urgency, fear, curiosity — to get someone to reveal information or take an action they otherwise wouldn't.

Common Tactics

- Phishing — fraudulent emails or messages impersonating a trusted sender.
- Pretexting — inventing a false scenario to extract information (e.g., posing as IT support).
- Baiting — offering something enticing (free download, USB drive) that carries malware.
- Vishing — phone-based social engineering.
- Tailgating — following an authorized person into a restricted area without their own credentials.

Spotting Phishing Red Flags

Warning signs include urgent or threatening language, mismatched sender addresses, generic greetings, unexpected attachments or links, and requests for credentials or payment. Hovering over a link (without clicking) to preview its real destination is a simple, effective habit.

Building a Human Firewall

A 'human firewall' refers to a well-trained workforce that recognizes and reports social engineering attempts rather than falling for them. Regular awareness training, simulated phishing tests, and a culture where people feel comfortable double-checking suspicious requests all strengthen this human layer of defense.



Figure 8.1 — Common Social Engineering Tactics

Key Terms

Social Engineering	**Phishing**
Pretexting	**Baiting**
Vishing	**Tailgating**
Human Firewall	**Security Awareness Training**

Review Questions

1. Why do attackers often target people instead of only technical systems?
2. List three red flags of a phishing email.
3. What is pretexting, and how does it differ from phishing?
4. What does it mean to build a 'human firewall'?
5. How can simulated phishing tests help an organization?

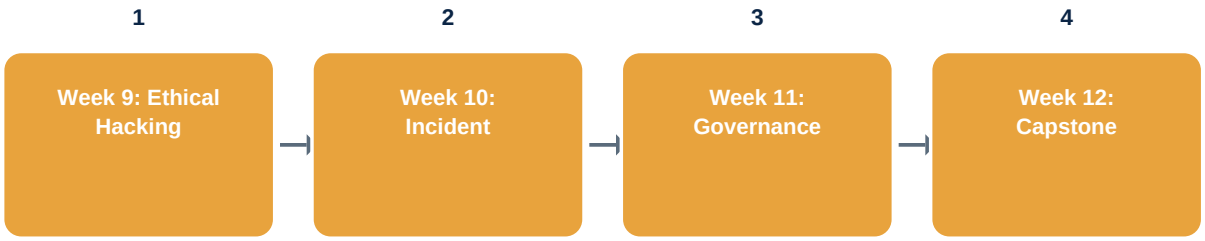
Practical Activity

- Collect three real (or publicly shared example) phishing messages and annotate the red flags in each.

- Draft a one-page phishing-awareness checklist for coworkers or classmates.

Month 3 — Practical & Career Readiness

Month 3 applies everything learned so far to real practice: how authorized security testing works, how organizations respond when incidents happen, how governance and compliance keep programs accountable, and how to turn this course into a career.



Week 9

Ethical Hacking & Penetration Testing Basics

Learning Objectives

- Explain the purpose and scope of authorized security testing.
- Describe the general penetration testing lifecycle.
- Understand the importance of authorization and rules of engagement.
- Explain responsible disclosure.

What Is Ethical Hacking?

Ethical hacking (also called penetration testing) is the practice of testing systems with proper authorization to find weaknesses before criminals do. The defining feature is permission: the same technical skills used maliciously by Black Hat hackers become a legitimate service when performed under a signed agreement, with a defined scope, for the system owner's benefit.

Types of Testing

- Black-box testing — the tester has no prior knowledge of the system, simulating an outside attacker.
- White-box testing — the tester has full knowledge and access, simulating an insider or a thorough audit.
- Gray-box testing — the tester has partial knowledge, a common middle ground.

Scope and Rules of Engagement

Before any testing begins, the organization and tester agree on scope (which systems may be tested), timing, methods that are allowed or off-limits, and emergency contact procedures. Testing outside the agreed scope — even with good intentions — is unauthorized access and can be illegal.

Responsible Disclosure

When a security researcher finds a vulnerability, responsible disclosure means privately reporting it to the affected organization first, giving them reasonable time to fix it before any public disclosure. Many companies run bug bounty programs that formalize and reward this process.

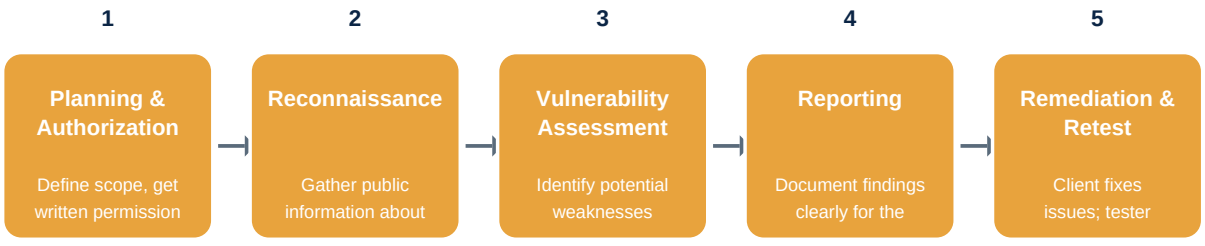


Figure 9.1 — Authorized Security Testing Lifecycle

Key Terms

Ethical Hacking	**Penetration Testing**
Black-Box Testing	**White-Box Testing**
Gray-Box Testing	**Scope**
Rules of Engagement	**Responsible Disclosure**
Bug Bounty	

Review Questions

1. What is the key difference between ethical hacking and illegal hacking?
2. Explain black-box vs. white-box testing.
3. Why is a defined scope critical before testing begins?
4. What is responsible disclosure, and why does it matter?
5. What is a bug bounty program?

Practical Activity

- Research one publicly known bug bounty program and summarize how it works.
- Draft a short (mock) authorization statement outlining scope and rules of engagement for a hypothetical test.

Week 10 Incident Response & Digital Forensics

Learning Objectives

- Outline the incident response lifecycle.
- Explain containment, eradication, and recovery.
- Understand chain-of-custody basics in digital forensics.
- Draft a simple incident report.

Why Incident Response Matters

No defense is perfect, so organizations prepare for the moment something goes wrong. Incident response (IR) is the structured process of detecting, containing, and recovering from security incidents while minimizing damage and downtime.

The Incident Response Lifecycle

A widely used model breaks IR into six phases: Preparation (policies, tools, training before anything happens), Identification (detecting and confirming an incident), Containment (limiting the damage, e.g., isolating an infected machine from the network), Eradication (removing the root cause), Recovery (restoring systems safely), and Lessons Learned (reviewing what happened to improve future response).

Digital Forensics Basics

Digital forensics involves collecting and analyzing evidence from computers or networks in a way that preserves its integrity, often because it may later be needed for legal or disciplinary action. Chain of custody refers to documenting exactly who accessed evidence, when, and how, so its authenticity can't be reasonably disputed later.

Writing an Incident Report

A good incident report is factual and chronological: what was detected, when, what actions were taken, what the root cause was, and what will change to prevent recurrence. Clear reporting helps both technical teams and leadership understand impact and response effectiveness.

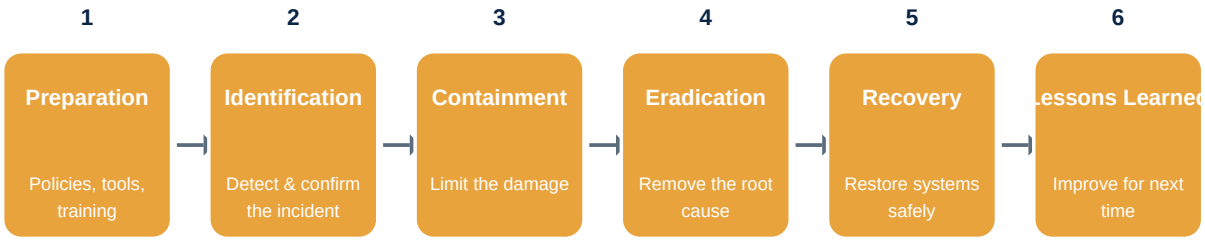


Figure 10.1 — The Incident Response Lifecycle (PICERL)

Key Terms

Incident Response	**Containment**
Eradication	**Recovery**
Digital Forensics	**Chain of Custody**
Root Cause Analysis	

Review Questions

1. List the six phases of the incident response lifecycle.
2. What is the goal of the containment phase?
3. Why is chain of custody important in digital forensics?
4. What should a good incident report include?
5. Why is the 'Lessons Learned' phase valuable even after recovery is complete?

Practical Activity

- Draft a one-page mock incident report for a fictional phishing-related account compromise.
- Research a well-known real-world data breach and summarize which IR phases were reported publicly.

Week 11

Security Compliance & Governance

Learning Objectives

- Explain why compliance frameworks exist.
- Differentiate policy, standard, and procedure.
- Describe basic risk-assessment steps.
- Recognize major frameworks (ISO 27001, NIST CSF) at an introductory level.

Why Governance Matters

Security governance is how an organization directs and controls its security efforts — setting expectations, assigning responsibility, and measuring whether protections are actually working. Compliance frameworks give organizations (and regulators, customers, and partners) a shared standard to measure against.

Policy, Standard, and Procedure

A policy is a high-level statement of intent (e.g., “All company data must be encrypted at rest”). A standard specifies the required implementation detail (e.g., “Use AES-256 encryption”). A procedure is the step-by-step instructions for how staff actually carry that out. Together they turn security goals into consistent, repeatable practice.

Introductory Frameworks

ISO/IEC 27001 is an international standard for building and maintaining an information security management system. The NIST Cybersecurity Framework (CSF) organizes security activities into functions — Identify, Protect, Detect, Respond, Recover — giving organizations a common language for maturity and gaps.

Basic Risk Assessment

Risk assessment generally involves identifying assets, identifying threats and vulnerabilities affecting them, estimating likelihood and impact, and deciding whether to accept, reduce, transfer, or avoid each risk. This prioritization helps organizations spend limited security budgets where they matter most.

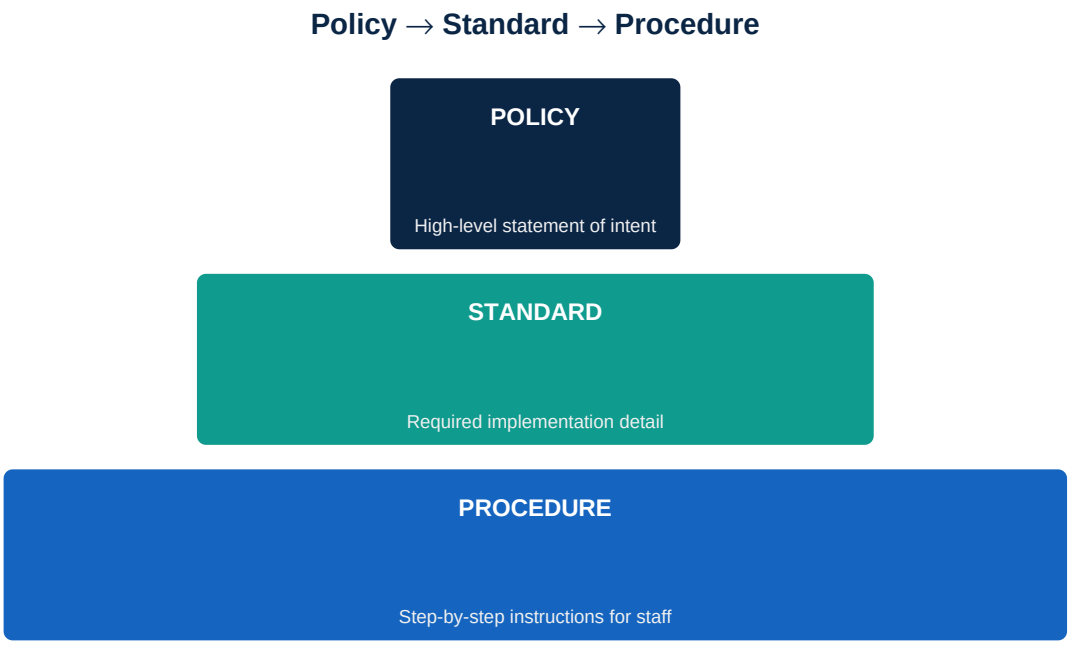


Figure 11.1 — Governance Hierarchy: Policy, Standard, Procedure

Key Terms

Governance	**Compliance**
Policy	**Standard**
Procedure	**ISO 27001**
NIST CSF	**Risk Assessment**

Review Questions

1. What is the difference between a policy, a standard, and a procedure?
2. Why do organizations pursue frameworks like ISO 27001?
3. Name the five functions of the NIST Cybersecurity Framework.
4. Outline the basic steps of a risk assessment.
5. What are the four common ways to respond to an identified risk?

Practical Activity

- Find one publicly available security policy (many universities and companies publish theirs) and identify its main components.
- Perform a mock risk assessment on a personal device: list two assets, one threat, and one way to reduce the risk.

Week 12

Capstone Project & Career Pathways

Learning Objectives

- Apply course concepts to a mock security assessment.
- Compare entry-level certification options.
- Identify realistic entry-level career roles.
- Build a personal 90-day post-course learning plan.

Capstone Project Brief

Working individually or in small groups, students select a small fictional organization (provided or self-defined) and produce a short security assessment covering: a CIA Triad review of its key assets, at least three plausible threats, one policy recommendation, and a basic incident response outline. The goal is to demonstrate connected thinking across the whole course, not just isolated facts.

Capstone Deliverables

- A one-to-two page written assessment.
- A short presentation (5–10 minutes) summarizing findings and recommendations.
- At least one diagram (network map, risk matrix, or response flow).

Certification Pathways

Common entry points after a fundamentals course include CompTIA Security+ (broad, vendor-neutral, widely recognized for entry-level roles) and ISC2 Certified in Cybersecurity (CC) (a free/low-cost entry credential from a major industry body). These validate foundational knowledge to employers and often unlock interviews that would otherwise require prior experience.

Entry-Level Career Roles

- SOC Analyst I — monitors alerts and escalates suspicious activity.
- IT Support / Help Desk — a common stepping stone into security-adjacent work.
- Security Intern — hands-on exposure inside a security team.
- GRC Analyst (Governance, Risk, Compliance) — for those drawn to Week 11's material.

Building a 90-Day Plan

Momentum after a course like this matters. A simple 90-day plan might dedicate the first month to studying for a chosen certification, the second month to hands-on practice (a home lab, guided exercises, or a capture-the-flag style learning platform), and the third month to networking and applying to entry-level roles or internships.

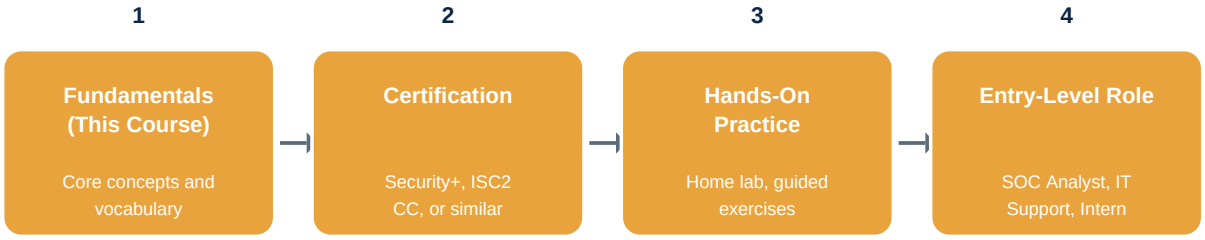


Figure 12.1 — Suggested Career Pathway After This Course

Key Terms

Capstone Project	**CompTIA Security+**
ISC2 Certified in Cybersecurity (CC)	**SOC Analyst**
GRC Analyst	**90-Day Plan**

Review Questions

1. What components should a capstone security assessment include?
2. Name two entry-level certifications discussed and what each is best suited for.
3. List two entry-level career roles in cybersecurity.
4. Why is hands-on practice recommended alongside certification study?
5. What should a well-structured 90-day post-course plan include?

Practical Activity

- Complete and present your capstone project security assessment.
- Write your own personal 90-day plan naming a target certification and one hands-on practice method.

Course Completion

■ Congratulations on completing all 12 weeks of the DDS Cybersecurity Fundamentals program. You've built a working vocabulary across security principles, networking, operating systems, cryptography, common threats, authorized testing, incident response, and governance — and closed with a capstone connecting it all together. The next step is practice: keep building, keep learning, and keep asking why a system is secure (or isn't).